Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 9545
CALIBRATION DATE: 17-Feb-19

Slocum Payload CTD CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -1.007882e+000
h =  1.460477e-001
i = -1.696171e-004
j =  3.210676e-005
```

```
CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC =  2.5356e-007
```

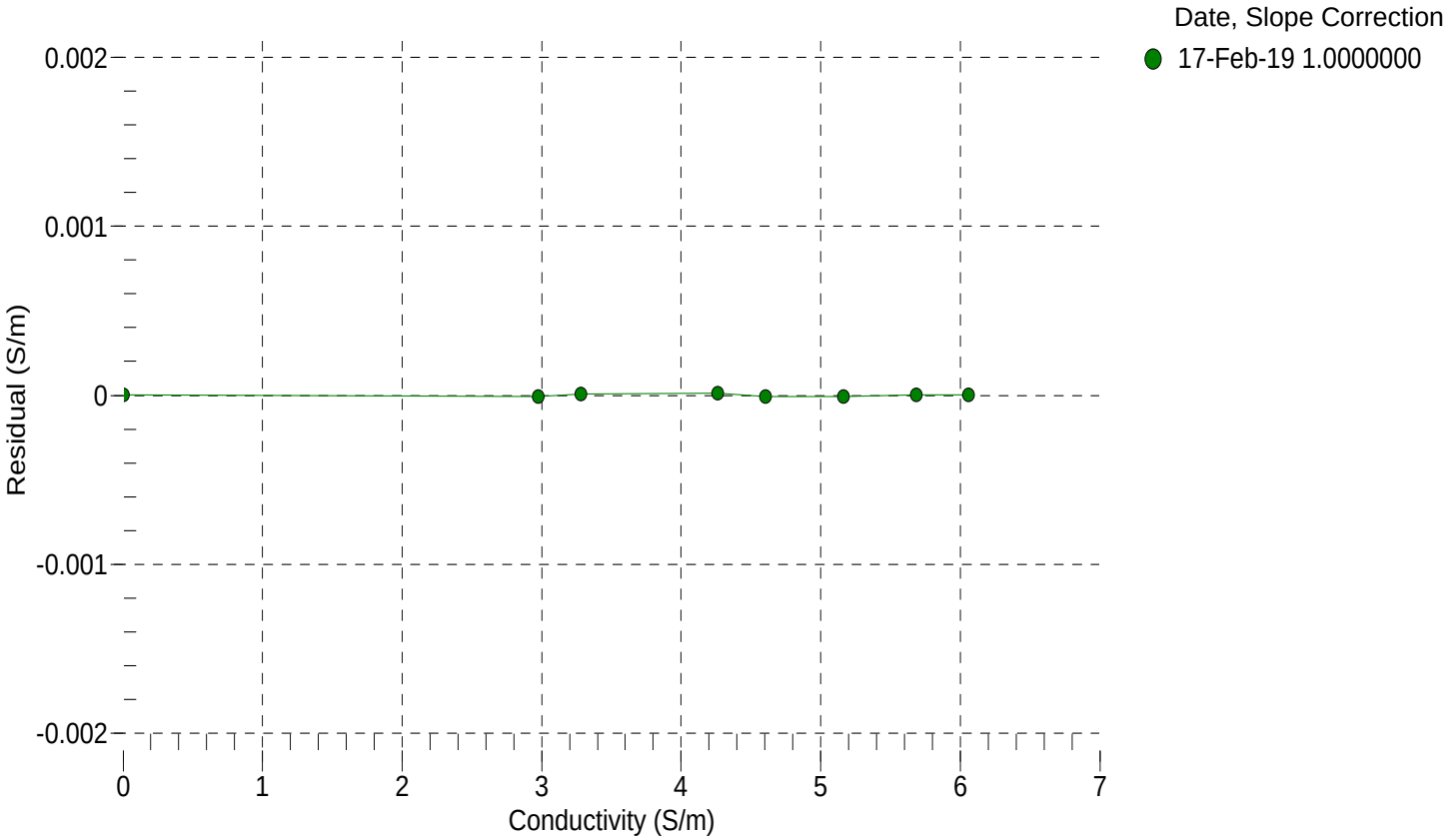
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2628.99	0.00000	0.00000
1.0000	34.7819	2.97330	5221.25	2.97329	-0.00001
4.5000	34.7629	3.28018	5418.15	3.28019	0.00001
15.0000	34.7224	4.26131	6003.80	4.26133	0.00001
18.5000	34.7143	4.60630	6196.34	4.60629	-0.00001
24.0000	34.7056	5.16398	6495.27	5.16397	-0.00001
29.0000	34.7012	5.68558	6762.65	5.68559	0.00000
32.5000	34.6982	6.05772	6946.95	6.05772	0.00000

f = Instrument Output(Hz) * sqrt(1.0 + WBOTC * t) / 1000.0

t = temperature (°C); p = pressure (decibars); δ = CTcor; ε = CPcor;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \varepsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity



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SENSOR SERIAL NUMBER: 9545
CALIBRATION DATE: 13-Feb-19

Slocum Payload CTD PRESSURE CALIBRATION DATA
1450 psia S/N 11151140

COEFFICIENTS:

```
PA0 =      -5.842182e-003
PA1 =       4.486184e-003
PA2 =      -6.891435e-012
PTEMPA0 = -6.515144e+001
PTEMPA1 =  5.203484e-002
PTEMPA2 = -6.049502e-007
```

```
PTCA0 =  5.237788e+005
PTCA1 =  9.887595e-001
PTCA2 = -4.866571e-002
PTCB0 =  2.508588e+001
PTCB1 = -2.250000e-004
PTCB2 =  0.000000e+000
```

PRESSURE SPAN CALIBRATION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)
14.42	526984.2	1721.6	14.39	-0.00
300.95	590856.1	1724.3	300.96	0.00
588.04	654829.0	1725.4	587.92	-0.01
875.45	718945.8	1725.9	875.48	0.00
1162.90	783009.1	1726.5	1162.74	-0.01
1450.04	847103.8	1725.5	1450.08	0.00
1162.77	783023.7	1725.3	1162.80	0.00
875.24	718919.8	1725.3	875.36	0.01
588.28	654910.5	1725.5	588.29	0.00
300.97	590868.6	1730.8	301.02	0.00
14.41	526992.8	1731.4	14.43	0.00

THERMAL CORRECTION

TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
32.50	1919	527038.80
29.00	1850	527044.80
24.00	1749	527048.60
18.50	1639	527054.60
15.00	1569	527066.20
4.50	1360	527061.40
1.00	1291	527055.60

TEMPERATURE (°C)	SPAN
-5.00	25.09
35.00	25.08

y = thermistor output (counts)

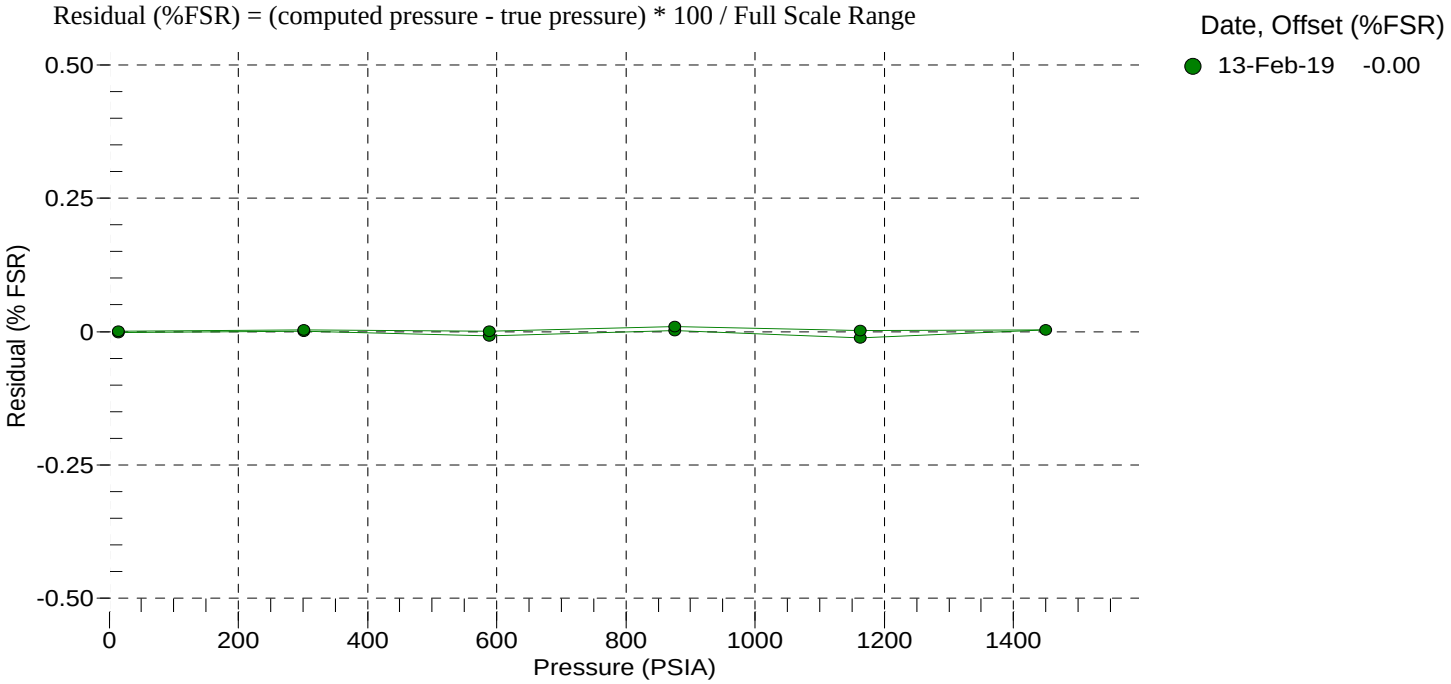
$t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y^2$

$x = \text{instrument output} - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$\text{pressure (PSIA)} = PA0 + PA1 * n + PA2 * n^2$

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range



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Pressure Test Certificate

Test Date: 2019-02-13

Description: Slocum CTD

Sensor Information:

Model Number: Slocum

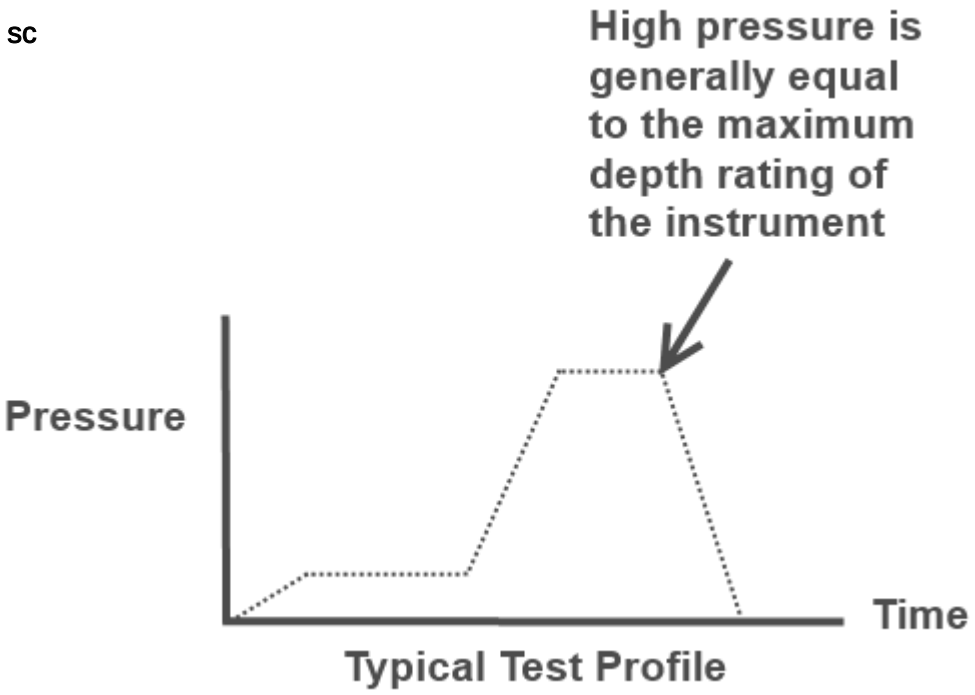
Serial Number: 9545

Pressure Test Protocol:

Low Pressure Test:	40	PSI	Held For:	15	Minutes
High Pressure Test:	1450	PSI	Held For:	15	Minutes

Passed Test: True

Tested By: sc



Typical Test Profile

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SENSOR SERIAL NUMBER: 9545
CALIBRATION DATE: 17-Feb-19

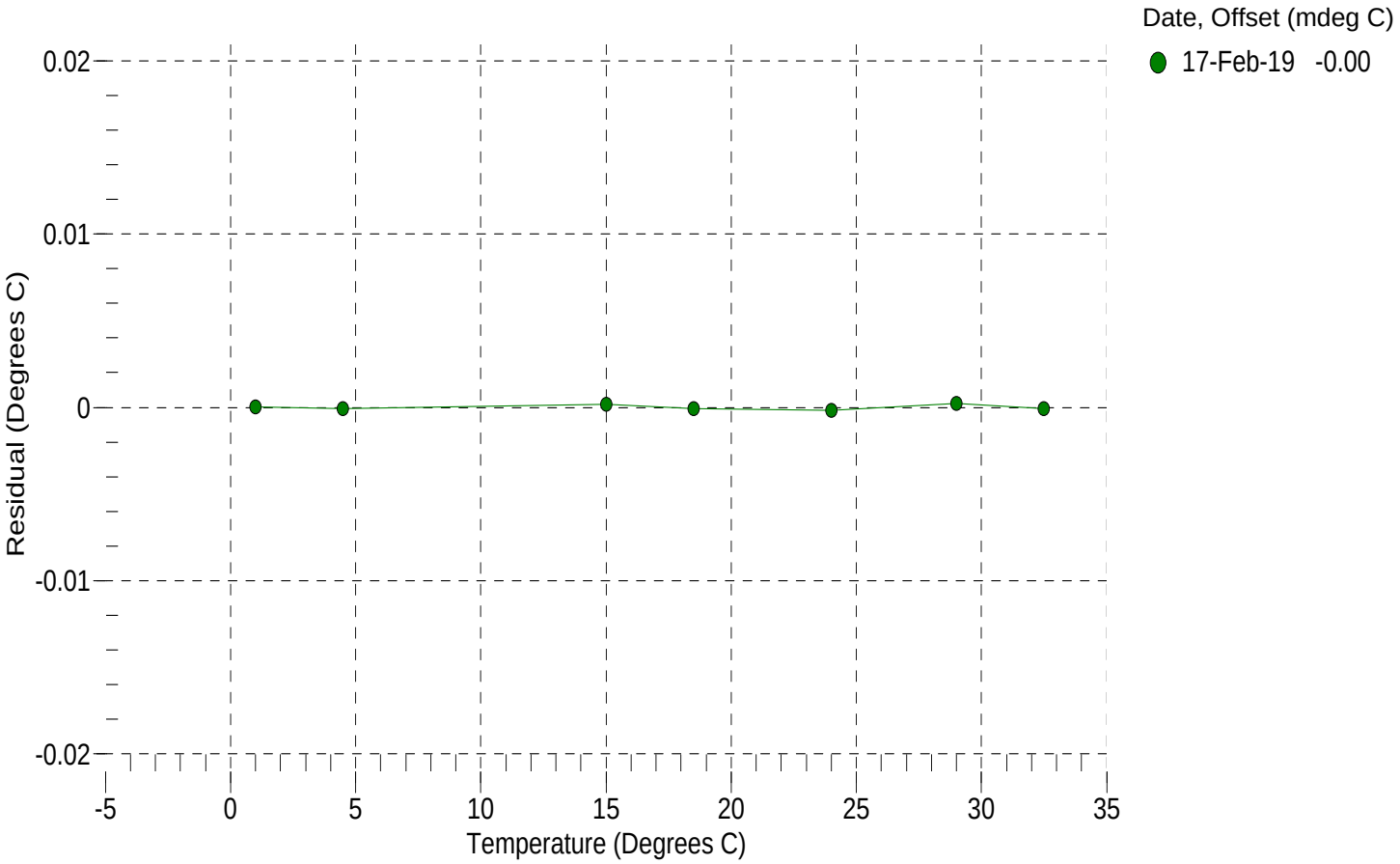
Slocum Payload CTD TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

```
a0 = -1.287019e-004
a1 =  3.068708e-004
a2 = -4.209381e-006
a3 =  1.932029e-007
```

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	567845.0	1.0000	0.0000
4.5000	486285.2	4.4999	-0.0001
15.0000	311441.8	15.0002	0.0002
18.5000	270136.0	18.4999	-0.0001
24.0000	217300.2	23.9998	-0.0002
29.0000	179377.6	29.0002	0.0002
32.5000	157368.2	32.4999	-0.0001

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature